

Mapping the quantum limits of classical simulatability

L. Shackleton, *Phys. Rev. Lett.* **136**, 186503 (2026)

L. Shackleton, *in progress*

Leyna Shackleton

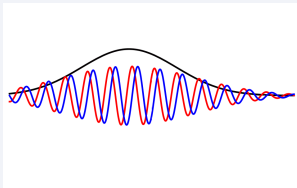
MIT

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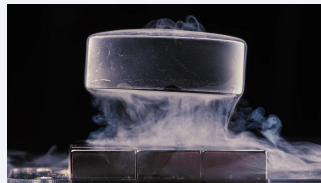
Emergent phenomena in many-body systems



many simple constituents



one quantum wavefunction

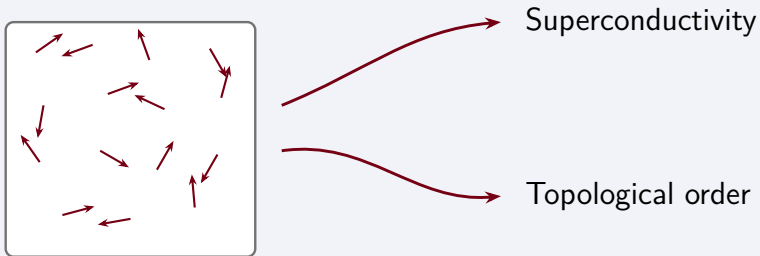


Superconductors
Quantum Hall phases
Quantum spin liquids
Topological order

...

How do we simulate quantum many-body systems?

Emergent phenomena in many-body systems



How do we simulate quantum many-body systems?

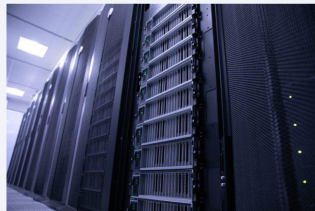
**Can some quantum phases of matter
be intrinsically hard to simulate?**

Simulating quantum systems is hard

Quantum Phase Transition at Nonzero Doping in a Random t - J Model

[Leyna Shackleton](#) ¹, [Alexander Wietek](#) ², [Antoine Georges](#) ^{2,3,4,5}, and [Subir Sachdev](#) ¹

The spectral function for the random t - J model, calculated using the Lanczos algorithm on an 18-site cluster, is shown for several values of doping in Fig. 1. A



Why are quantum systems hard to simulate?



$|\uparrow\rangle$

$|\downarrow\rangle$

1 particle

$|\uparrow\uparrow\rangle$

$|\uparrow\downarrow\rangle$

$|\downarrow\uparrow\rangle$

$|\downarrow\downarrow\rangle$

2 particles

$|\uparrow\uparrow\uparrow\rangle$

$|\uparrow\uparrow\downarrow\rangle$

$|\uparrow\downarrow\uparrow\rangle$

$|\uparrow\downarrow\downarrow\rangle$

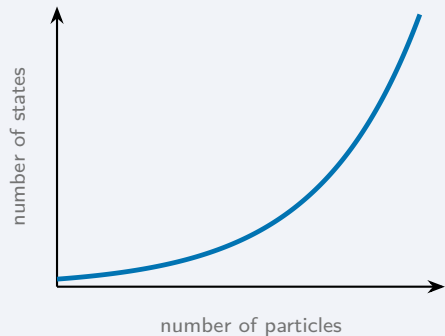
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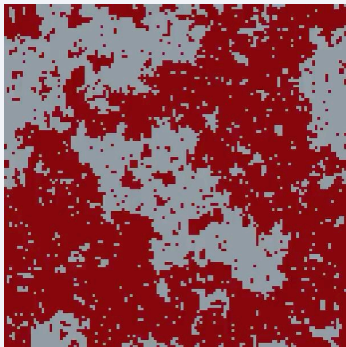
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3 particles



Simulating classical systems is (more) easy

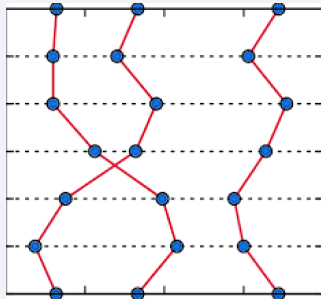


$$Z = \sum_{\{\sigma_i\}} e^{-E(\{\sigma_i\})/k_B T}$$
$$\langle O \rangle = \frac{\sum_{\{\sigma_i\}} O(\{\sigma_i\}) e^{-E(\{\sigma_i\})/k_B T}}{Z}$$

exponential sum!

Amenable to Monte Carlo sampling

Quantum obstacle: the sign problem

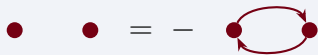


$$Z = \text{Tr} \left[e^{-\beta \hat{H}} \right]$$

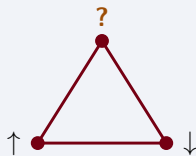
$$\langle \hat{O} \rangle = \frac{\text{Tr} \left[\hat{O} e^{-\beta \hat{H}} \right]}{Z}$$

Weights are **no longer positive**

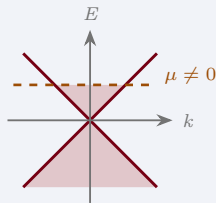
The sign problem is ubiquitous for quantum systems



Fermions

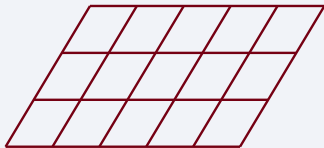


Frustration



Finite density

Sign problem-free models provide invaluable insight



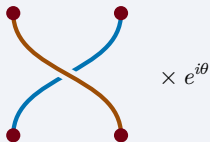
Lattice QCD

Quantum magnetism

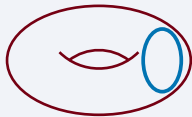
Quantum criticality

**Can a phase of matter have
an intrinsic sign problem?**

“Intrinsic” sign problems for topological phases



braiding anyons imprints a phase



ground-state degeneracy set by topology

- Double semion topological order
(Hastings 2016)
- Most topological orders
(Ringel 2017, Seo 2025)
- “We analyze the Gauss sums of 405 different topological orders classified up to rank 12 [...] and find that 398 suffer from the intrinsic sign problem.”

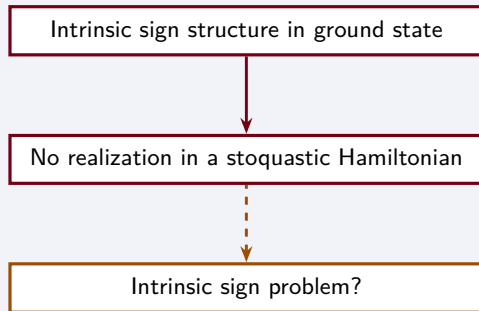
Stoquasticity is a flawed measure of the sign problem

$$Z = \text{Tr} \left[e^{-\beta \hat{H}} \right]$$

↑ completely positive matrix: “stoquastic”

Perron–Frobenius

A stoquastic Hamiltonian must have a completely positive ground state.

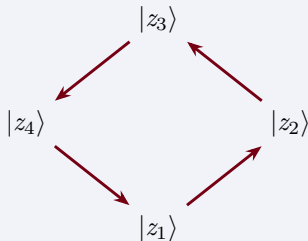


This last step does not follow.

How does the sign problem appear?

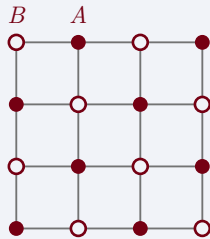
$$Z = \sum_j \sum_n \sum_{\{b_1, \dots, b_n\}} \frac{(-\beta)^n}{n!} \langle z_j | H_{b_n} \cdots H_{b_2} H_{b_1} | z_j \rangle$$

$$H = \sum_b H_b$$



- Stoquasticity a **sufficient** but **not necessary** condition
- More general: **global cancellation of phases**
- Sign structure invariant under $|z_j\rangle \rightarrow e^{i\theta_j} |z_j\rangle$
- Sign problem-free if it can be made stoquastic via many-body phase rotations
(Hen 2021)

Well-known example in quantum magnetism: the Heisenberg antiferromagnet



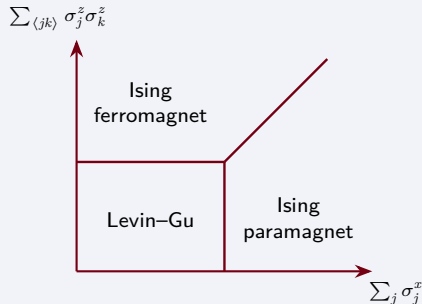
bipartite lattice

$$H = J \sum_{\langle jk \rangle} \mathbf{S}_j \cdot \mathbf{S}_k, \quad J > 0$$

- Off-diagonal terms $\frac{J}{2}(S_j^+ S_k^- + S_j^- S_k^+)$ are **positive** — not stoquastic.
- **Marshall sign rule.** Rotate every spin on one sublattice, $U = \prod_{j \in B} e^{i\pi S_j^z}$.
- $UHU^\dagger$ is **stoquastic**: a many-body phase rotation removes the sign problem entirely.

New example: the Levin–Gu SPT

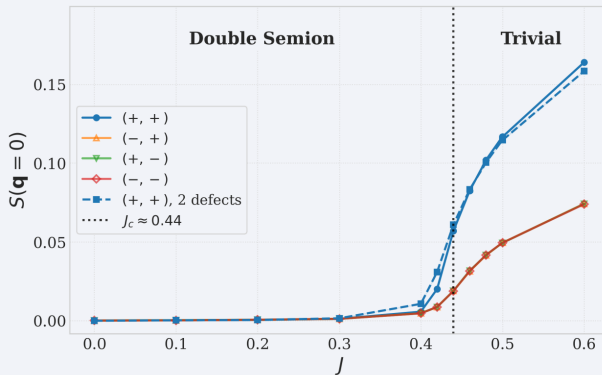
$$H_{\text{Levin-Gu}} = \sum_j \sigma_j^x \prod_{\triangle_{jkl}} i^{\frac{1}{2}(-1-\sigma_k^z \sigma_l^z)} \equiv - \sum_j B_j$$



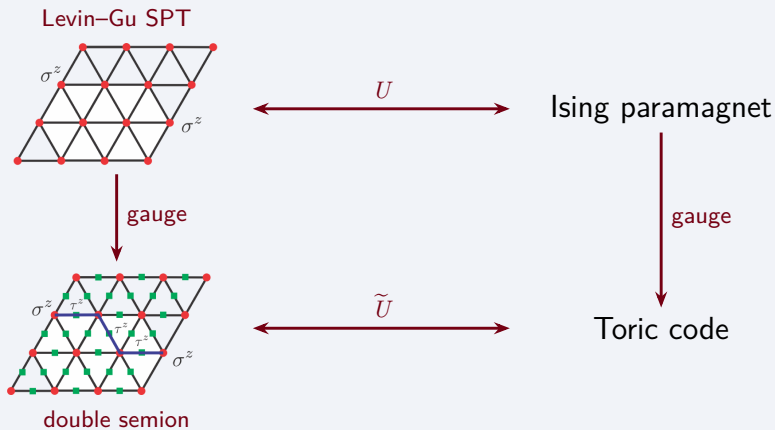
- $\mathbb{Z}_2$ SPT in $(2+1)\text{D}$, hosts a symmetry-protected sign problem (Ellison 2021)
- Domain wall parity changes an **even** number of times in SSE — net sign is $+1$
- Related to the Ising paramagnet by $U = (-1)^{\#DW}$
- Sign structure preserved under σ^x and σ^z perturbations

The double semion model: sign problem-free

- Not fine-tuned
- No sophisticated tricks — uses conventional QMC methods
- Hamiltonians are standard and well-known

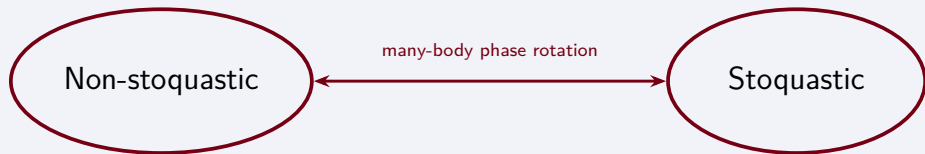


Cancellation mechanism via connection to SPT



SSE sign structures of H_{TC} and H_{DS} are identical.

What are obstacles to stoquasticizability?



- Refined criterion: sign problem-free models are **stoquasticizable**
- Are there obstacles to stoquasticizability?
- Chiral phases are an obstruction!

References

- L. Shackleton, *Twisted quantum doubles are sign problem-free*, Phys. Rev. Lett. **136**, 186503 (2026).
- L. Shackleton and S. Sachdev, Phys. Rev. B **111**, 075101 (2025).
- M. B. Hastings, *How quantum are non-negative wavefunctions?*, J. Math. Phys. **57**, 015210 (2016).
- Z. Ringel and D. L. Kovrizhin, *Quantized gravitational responses, the sign problem, and quantum complexity*, Sci. Adv. **3**, e1701758 (2017).
- T. D. Ellison, K. Kato, Z.-W. Liu and T. H. Hsieh, *Symmetry-protected sign problem and magic in quantum phases of matter*, Quantum **5**, 612 (2021).
- I. Hen, *Determining QMC simulability with geometric phases*, Phys. Rev. Research **3**, 023080 (2021).
- X. Chen, Z.-C. Gu and X.-G. Wen, Phys. Rev. B **84**, 235128 (2011).
- M. Levin and Z.-C. Gu, Phys. Rev. B **86**, 115109 (2012).

Thank you